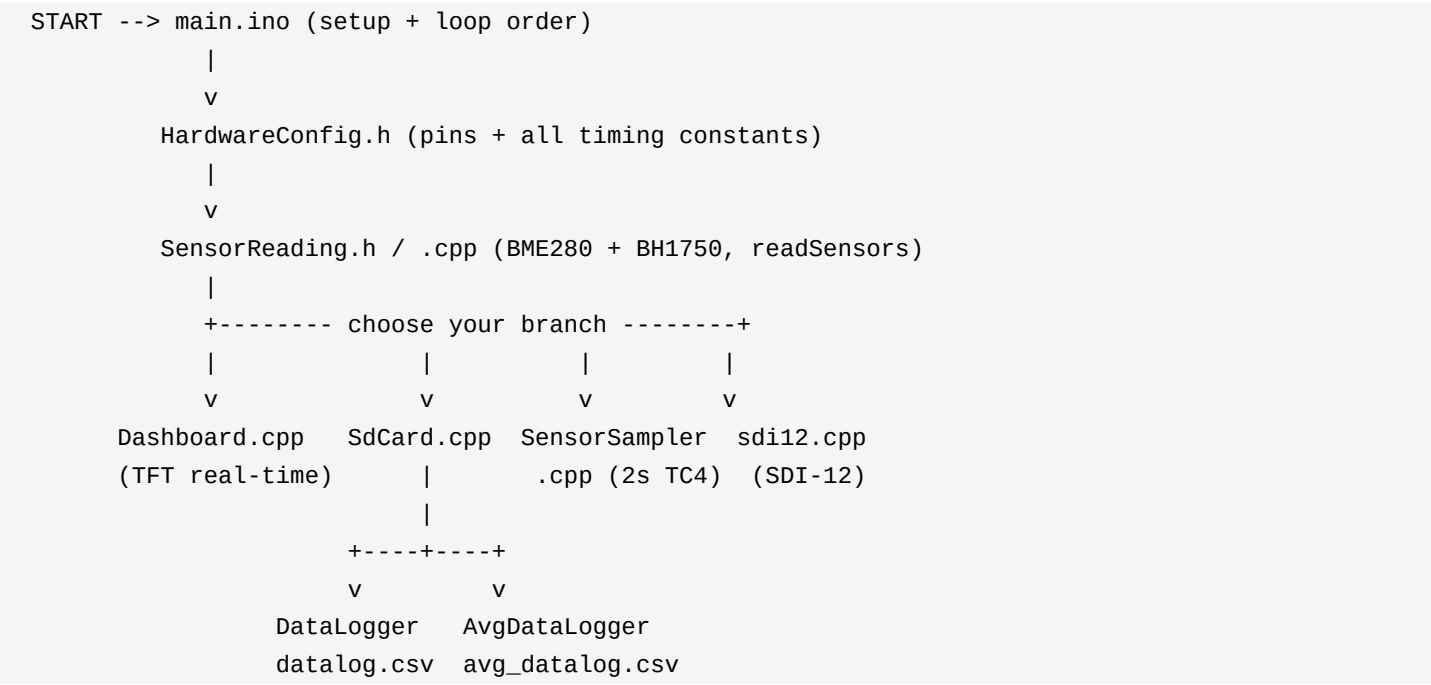


SDI-12 Sensor Station - System Workflow

Board: Arduino Zero / MKR / SAMD21 (TC4 hardware timer). This document explains how main.ino and all modules work together.

0. Where to START reading the code

Follow this order the first time you open the project:

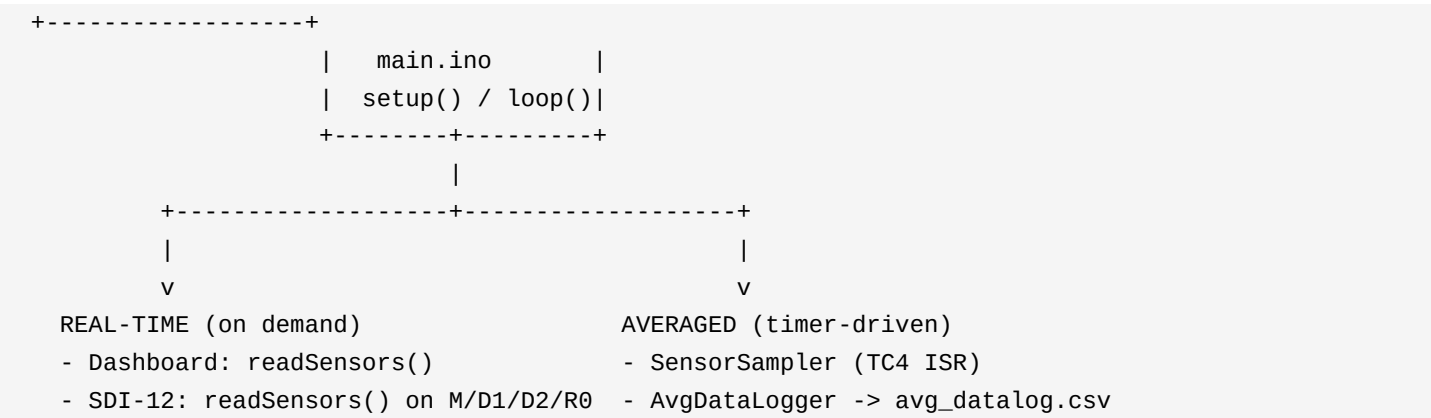


Step	File	Why
1	main.ino	Only entry point - what runs and in what order
2	HardwareConfig.h	Pins and intervals (2s, 60s, 1s)
3	SensorReading.cpp	All sensors - every module uses this
4a	Dashboard.cpp	TFT display (real-time reads)
4b	SdCard + DataLogger	datalog.csv + buttons
4c	SensorSampler + AvgDataLogger	2s averages + avg_datalog.csv
4d	sdi12.cpp	SDI-12 commands M, D1, D2, R0

Rule: Always read main.ino -> HardwareConfig.h -> SensorReading first. Then open only the branch you need.

1. Big picture (two separate sensor paths)

- REAL-TIME path: instant I2C reads for TFT, SDI-12, and datalog.csv
- AVERAGED path: TC4 timer every 2 s -> avg_datalog.csv only



- DataLogger: readSensors() - Every 2 seconds
- > datalog.csv every 60 s

2. Module map (what each file does)

File	Role
main.ino	Entry point. Calls all init and loop handlers in order.
HardwareConfig.h	Pins, intervals: dashboard 1s, datalog 60s, sample 2s.
SensorReading.cpp	BME280 + BH1750 I2C. readSensors(), getSensorData().
SensorSampler.cpp	TC4 ISR (10ms base, 2s sample). Builds 2s averages.
SdCard.cpp	One SD card + RTC shared by both loggers.
DataLogger.cpp	datalog.csv - periodic (60s) + manual button.
AvgDataLogger.cpp	avg_datalog.csv - one row per 2s average.
Dashboard.cpp	TFT display - real-time readSensors(), NOT averages.
sdi12.cpp	SDI-12 slave on Serial1. Uses getSensorData() after read.

3. setup() order (runs once at power-on)

```

Serial.begin(9600)
  sensorsInit()           // Wire + BME280 + BH1750
  sdCardInit()            // RTC + SD card (shared)
  sensorSamplerInit()     // Start TC4 timer (SAMD21)
  dataLoggerInit()        // Buttons + datalog.csv header
  avgDataLoggerInit()     // avg_datalog.csv header
  dashboardInit()         // TFT labels
  sdi12Init('0', pin 8)   // SDI-12 bus
  readSensors()           // First reading

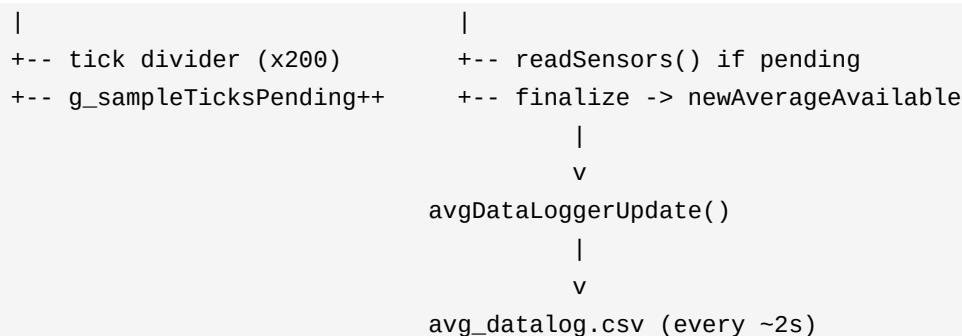
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4. loop() order (runs forever)

1. sensorSamplerService() // FIRST: process 2s timer flags, I2C for averages
2. sdi12Handle() // SDI-12 commands + R0 stream (1s)
3. avgDataLoggerUpdate() // Write avg_datalog.csv when new 2s average ready
4. dataLoggerUpdate() // 60s log + manual/clear buttons
5. dashboardUpdate() // TFT refresh every 1s, real-time readSensors()

5. Averaged path detail (SensorSampler)

- TC4 hardware interrupt fires every ~10 ms (base tick).
- ISR counts 200 ticks -> one sample request every 2000 ms (2 s).
- ISR only sets g_sampleTicksPending (NO I2C inside interrupt).
- sensorSamplerService() in loop reads sensors and accumulates.
- After 1 sample in window -> finalizeAverageWindow().
- AvgDataLogger writes row and calls sensorSamplerAcknowledgeAverage().



6. Real-time path detail (everything else)

- Dashboard: every 1000 ms calls readSensors() then getSensorData().
- DataLogger: every 60000 ms OR button pin 2 -> datalog.csv row.
- SDI-12: host sends command ending with ! -> readSensors() -> reply.
- SDI-12 R0 stream: every 1000 ms while streaming -> readSensors().
- Clear SD button (pin 3): deletes datalog.csv only (not avg file).

7. CSV files (do not mix them up)

File	Interval	Data type
datalog.csv	60 s + manual	Instant snapshot at log time
avg_datalog.csv	~2 s	Hardware-timed average (1 sample/window)

8. SDI-12 commands (address 0 by default)

- ? or 0! -> Respond with sensor address
- 0M! -> Measure: returns count of parameters
- 0D1! -> Send BME: temp, humidity, pressure
- 0D2! -> Send BH1750: lux
- 0R0! -> Send all + start continuous stream every 1 s
- 0An! -> Change address to n

9. Timing constants (HardwareConfig.h)

- kSensorSampleMs = 2000 ms (Average path: time between I2C samples)
- kSensorAverageMs = 2000 ms (Length of one average window)
- kTc4BasePeriodMs = 10 ms (TC4 hardware tick (divider -> 2 s))
- kDashboardRefreshMs = 1000 ms (TFT update rate)
- kLogIntervalMs = 60000 ms (datalog.csv auto-log interval)
- R0_STREAM_INTERVAL_MS = 1000 ms (SDI-12 R0 stream (in sdi12.cpp))

10. Common confusion - quick answers

Q: Why two sensor reads?

TC4

A: San

Q: Does TFT show averages?

A: No.

Q: Does SDI-12 use averages?

A: No.

Q: What does TC4 do?

A: Onl

Q: Non-SAMD board?

A: Ser

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